Methods and Models for Combinatorial Optimization

Lab exercise - Part I

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Consider the following combinatorial optimization problem and the proposed Integer Linear Programming (ILP) model. **The task is to *adapt the ILP model to the given problem*, to implement the model using the Cplex API or DoCplex or any other approved tool (see the table presented in class), and to test it**. Test should determine the ability of the model to provide exact solutions for the proposed problem: in particular we want to know up to which size (namely the number of holes) the problem can be solved in different amounts of time (up to 0.1 seconds, up to 1 second, up to 10 seconds ...).

1 Problem description

A company manufactures boards with holes that are used to assemble electrical panels (see Figure 1). Each board is placed on a machine, where a drill moves across the board, stopping at the required positions to drill the holes. Once a board has been fully processed, it is replaced with a new one, and the procedure is repeated. The drill starts from a home position, where no hole needs to be drilled, and returns to the same position once the board has been fully processed. The drilling process generates heat in the area surrounding each hole. For quality reasons, the drill can operate at a given position only if the surrounding area is sufficiently "cold". More specifically, a hole can be drilled at a given position only

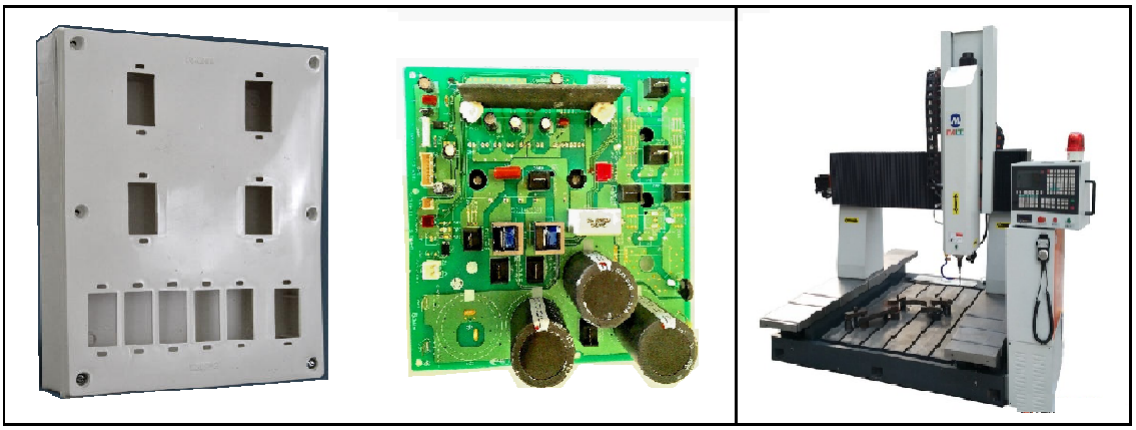


Figure 1: Sample boards for electric panels and an automatic drilling machine

if at least a specified number of holes have been drilled at positions sufficiently far from it since the last time a hole was drilled in its vicinity. Given the positions of the holes on the board, the threshold distance defining which positions are considered sufficiently far apart, and the minimum number of intermediate holes that must satisfy this distance requirement, the company asks us to determine the sequence in which the holes should be drilled so as to minimize the total drilling time, assuming that the time required to drill each hole is constant and identical for all holes.

2 Mathematical formulation

2.1 Modelling framework

We can represent the problem on a weighted complete graph $G = (N \cup \{0\}, A)$, where 0 is the home position, N is the set of nodes and corresponds to the set of holes that have to be drilled (one node for each hole), and A is the set of the arcs (i, j), $\forall\ i, j \in N \cup \{0\}$, representing the trajectory of the drill moving from the position of hole i to the position of hole j. A weight c_{ij} can be associated to each arc $(i, j) \in A$, corresponding to the time needed to move from i to j. In this graph model, the problem can be seen as finding the path of minimum weight that visits all the nodes. Indeed, since the drill has to come back to the initial home position in order to start with the next board, the path has to be a cycle. **If we neglect the requirement related to board heating,** the problem can be seen as determining the minimum weight hamiltonian cycle on G, that is, as a Travelling Salesman Problem (TSP) on G.

2.2 Mathematical Programming models

Several formulations for the TSP have been formulated based on (Mixed) Integer Linear Programming. Some of them include an exponential number of constraints (see Lecture Notes on "Exact methods for TSP") and, in practice, they they require row generation procedures and hence cannot be directly implemented using basic Solvers' APIs. We then focus on **compact formulations**, i.e., formulations that requires a polynomial (possibly with small degree) number of variables and constraints.

In the following, we suggest a possible compact formulation for TSP based on network flows. It is not the only available compact formulation: **you may search the literature for an alternative compact formulation and consider the opportunity of implementing it (this would be appreciated, in particular if you do not use Cplex C-APIs for implementation**).

Warning! We recall that the exercise problem includes requirements related to board heating, hence the model must be adapted before implementation.

2.3 A base model for TSP based on network flows

The TSP can be formulated (among others) as a network flow model on G. Let $|N|$ be the amount of flow outgoing from node 0. The idea is to push this amount of flow towards

the remaining nodes in such a way that (i) each node (different from 0) receives 1 unit of flow, (ii) each node is visited once, and (iii) the sum of c_{ij} over all the arcs shipping some flow is minimum.

Sets:

- N = graph nodes representing the holes;
- A = arcs (i, j), $\forall\ i, j \in N \cup \{0\}$, representing the trajectory covered by the drill to move from hole i to hole j.

Parameters:

- c_{ij} = time taken by the drill to move from i to j, $\forall\ (i, j) \in A$.

Decision variables:

- x_{ij}= amount of the flow shipped from i to j, $\forall\ (i, j) \in A$;
- y_{ij}= 1 if arc (i, j) ships some flow, 0 otherwise, $\forall\ (i, j) \in A$.

Base Integer Linear Programming model for TSP:

$$
\begin{aligned}
\min \quad & \sum_{i,j:(i,j)\in A} c_{ij} y_{ij} && && && (1)\\
s.t. \quad & \sum_{j:(0,j)\in A} x_{0j} && = && |N| && (2)\\
& \sum_{i:(i,k)\in A} x_{ik} - \sum_{j:(k,j)} x_{kj} && = && 1 \qquad \forall\ k \in N && (3)\\
& \sum_{j:(i,j)\in A} y_{ij} && = && 1 \qquad \forall\ i \in N \cup \{0\} && (4)\\
& \sum_{i:(i,j)\in A} y_{ij} && = && 1 \qquad \forall\ j \in N \cup \{0\} && (5)\\
& x_{ij} && \leq && |N|\ y_{ij} \quad \forall\ (i, j) \in A && (6)\\
& x_{ij} \in \mathbb{R}_+ && && \forall\ (i, j) \in A && (7)\\
& y_{ij} \in \{0, 1\} && && \forall\ (i, j) \in A && (8)
\end{aligned}
$$

We notice that the constraint (2) is redundant: by (3), at least $|N|$ units of flows must be available; moreover, by optimality, there is no need to send flow towards node 0, so that we can fix $x_{i0} = 0$, for all $i \in N$, i.e., we can eliminate those variables. We thus obtain

the following formulation.

$$\min \quad \sum_{i,j:(i,j)\in A} c_{ij} y_{ij} \tag{9}$$

$$s.t. \quad \sum_{i:(i,k)\in A} x_{ik} - \sum_{j:(k,j),j\neq 0} x_{kj} = 1 \qquad \forall\, k \in N \tag{10}$$

$$\sum_{j:(i,j)\in A} y_{ij} = 1 \qquad \forall\, i \in N \cup \{0\} \tag{11}$$

$$\sum_{i:(i,j)\in A} y_{ij} = 1 \qquad \forall\, j \in N \cup \{0\} \tag{12}$$

$$x_{ij} \leq (|N|)\ y_{ij} \qquad \forall\, (i,j) \in A, j \neq 0 \tag{13}$$

$$x_{ij} \in \mathbb{R}_+ \qquad \forall\, (i,j) \in A, j \neq 0 \tag{14}$$

$$y_{ij} \in \{0,1\} \qquad \forall\, (i,j) \in A \tag{15}$$

corresponding to the one presented in the paper

> Gavish B., Graves S. (1978), The travelling salesman problem and related problems, working Paper GR-078-78. Operations Research Center, Massachusetts Institute of Technology, Cambridge.

2.4 Adapting the base formulation

Before implementing the above formulation, it must be adapted to integrate the requirements related to board heating.

Hint: add suitable constraints based on observing that the quantity

$$\phi(k) = \sum_{j:(i,k)} x_{ik} \qquad \forall\, k \in N \tag{16}$$

represents the amount of flow entering node k and can be used to retrieve the order holes are processed by the solution. For example, if $N = \{a, b, c, d, e\}$ and the solution is the sequence $< 0, c, e, a, d, b, 0 >$, we will have $\phi(c) = 5$, $\phi(e) = 4$, $\phi(a) = 3$, $\phi(d) = 2$, $\phi(b) = 1$.